Reinforcement Learning Control

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1 Introduction

2 Simple Drive with Feedback Control

The fist application in which reinforcement learning is utilized to control a system is to control a simple second order point mass in the xy-plane. The dynamics of this system are given in Equation 1, where the state is defined by the xy-position and xy-velocity: $\boldsymbol{x} = [p_x, p_y, V_x, V_y]^T$ and the control input is the xy-acceleration: $\boldsymbol{u} = [a_x, a_y]^T$.

$$\begin{bmatrix} p_{x_{i+1}} \\ p_{y_{i+1}} \\ V_{x_{i+1}} \\ V_{y_{i+1}} \end{bmatrix} = \begin{bmatrix} 1 & 0 & dt & 0 \\ 0 & 1 & 0 & dt \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_{x_i} \\ p_{y_i} \\ V_{x_i} \\ V_{y_i} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ dt & 0 \\ 0 & dt \end{bmatrix} \begin{bmatrix} a_{x_i} \\ a_{y_i} \end{bmatrix} \tag{1}$$

In this application, a reinforcement learning (RL) policy is trained to find the gains of a feedback controller, given in Equation 2, where the velocity references and acceleration inputs are clipped at $5[m/s]$ and $2[m/s^2]$, respectively.

$$V_{x_{ref}} = Kp_x(p_{x_{ref}} - p_x) \qquad V_{y_{ref}} = Kp_y(p_{y_{ref}} - p_y) \tag{2}$$

$$a_x = kp_{V_x}(V_{x_{ref}} - V_x) \qquad a_y = kp_{V_y}(V_{y_{ref}} - V_y) \tag{3}$$

The RL policy is set up such that it provides the four controller gains based on a selected maximum overshoot percentage on the x- and y-position. It is trained to handle an overshoot of 10%, 15% and 20%. This is achieved through subtracting points from its reward if the trajectory crossed the specified overshoot in a given simulation run. Furthermore, tracking of the positional reference p_{ref} is encouraged through adding a cost, given in Equation 4, to the reward.

$$cost = -\sum_{i=0}^{T} \left((p_{ref} - p_i)^T Q (p_{ref} - p_i) + u_i^T R u_i \right) dt \tag{4}$$

The policy is trained using the *PPO* algorithm with a total of 4096 simulation runs. The results of the trained policy for an specified maximum overshoot of 10%, 15% and 20% are given in Figure 1, Figure 2 and Figure 3, respectively. From these results, it can be seen that the algorithm finds more agressive gains for a higher allowed maximum overshoot.

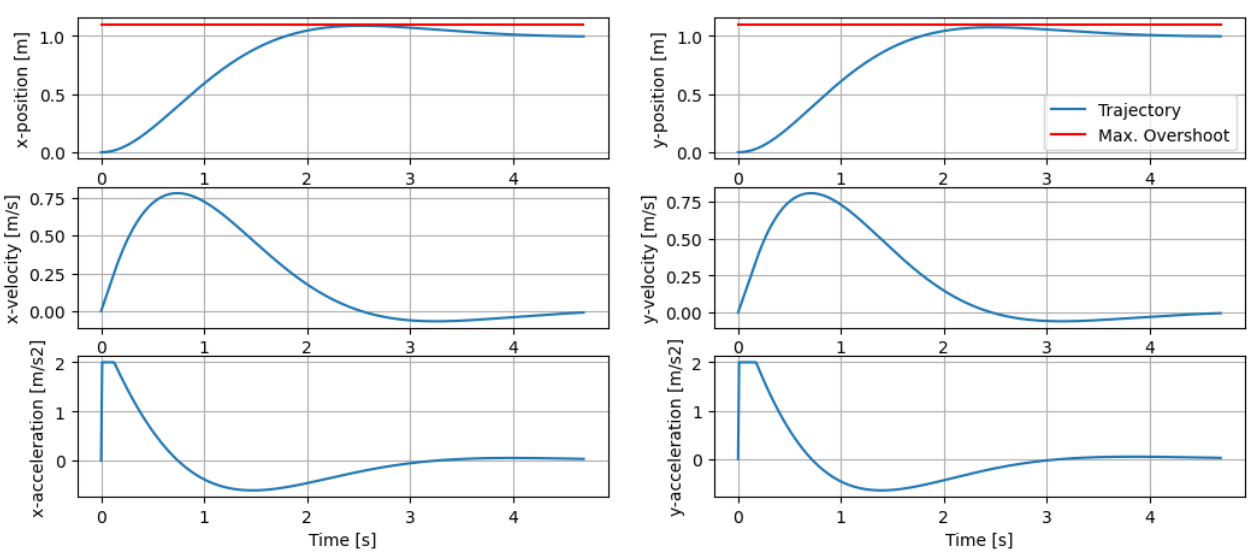

Figure 1: Trajectories for a maximum overshoot of 10% resulting in the following gains: $Kp_x = 1.27$, $Kp_y = 1.29$, $Kp_{Vx} = 1.96$ and $Kp_{Vy} = 2.15$

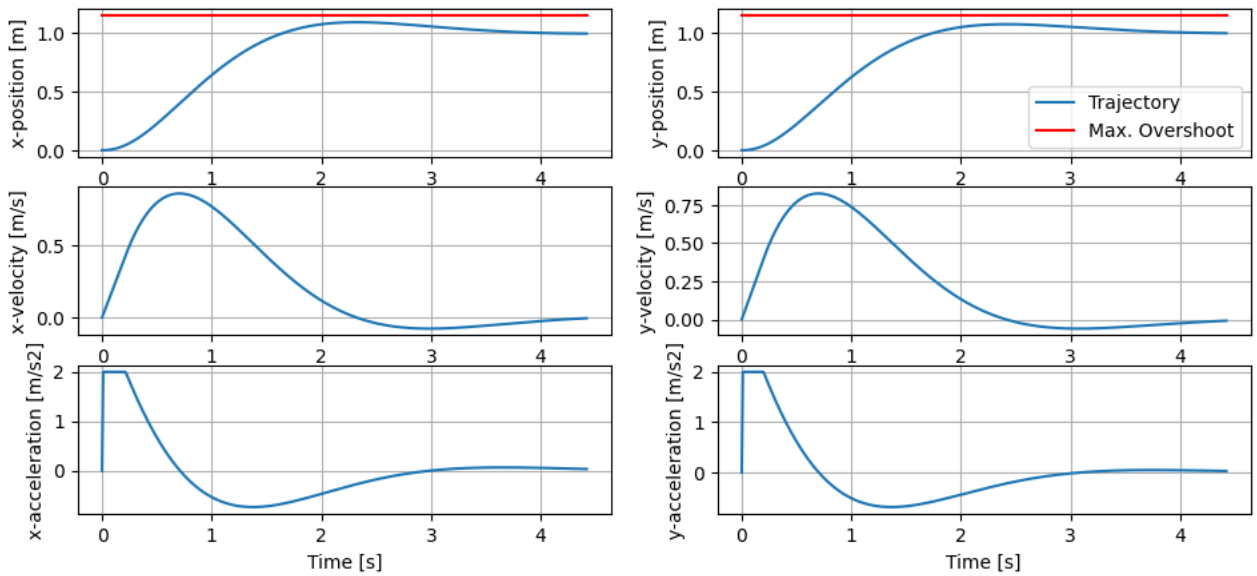

Figure 2: Trajectories for a maximum overshoot of 15% resulting in the following gains: $Kp_x = 1.41$, $Kp_y = 1.33$, $Kp_{Vx} = 2.12$ and $Kp_{Vy} = 2.21$

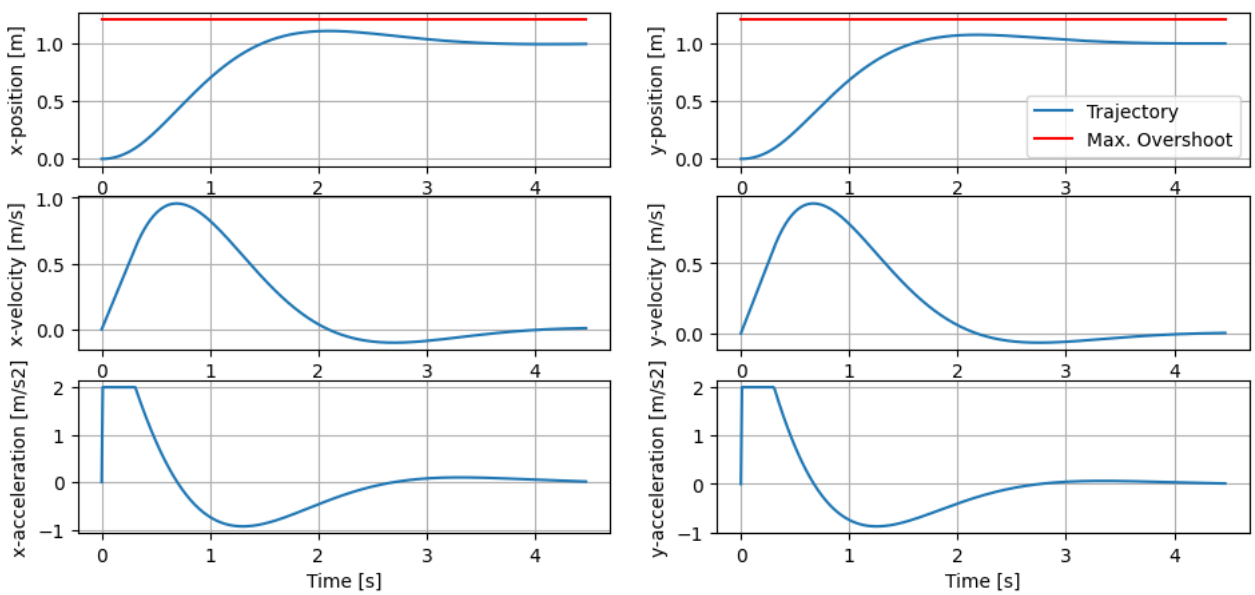

Figure 3: Trajectories for a maximum overshoot of 20% resulting in the following gains: $Kp_x = 1.63$, $Kp_y = 1.51$, $Kp_{Vx} = 2.26$ and $Kp_{Vy} = 2.54$

3 Differential Drive with Feedback Control